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# Visualization of cardiac health using vector cardiogram

# Visualisation de la santé cardiaque par le vecteur de cardiogramme

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**Abstract**

An electrocardiogram (ECG) is an electrical recording of the heart and is used in the investigation of heart disease. It displays an apparent periodicity (of about 60–100 bpm in a healthy adult), but is not exactly periodic. The symptoms of disease may show up only during certain periods of the day, and that too may occur at random in the time scale. Visual media is a most effective tool for communication, especially when the data has subtle variations. A novel visualization technique is presented to display each ECG beat. The features like PR interval, QRS width, ST interval, are extracted from the magnitude and phase plot of different lead combinations. These features are displayed on a Cartesian quadrant as different curves, with a menu driven display strategy to visualize the ECG for a chosen interval. The scheme employed can be used to identify different types of abnormalities.
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**Résumé**

Un électrocardiogramme est en quelque sorte « un enregistrement électrique » du cœur ; il est utilisé pour la détection des maladies cardiaques. L'électrocardiogramme (ECG) montre une périodicité apparente mais il n'est pas exactement périodique. Les symptômes de la maladie peuvent apparaître durant certaines périodes de la journée mais ils peuvent aussi se manifester de façon aléatoire dans le temps. Le moyen visuel est l'outil de communication le plus efficace particulièrement dans le cas de données pour lesquelles les variations sont subtiles. Une nouvelle visualisation technique montrant chaque battement de l'ECG est présentée. Les paramètres caractéristiques tels que l'intervalle PR, la largeur QRS, l'intervalle ST sont extraits à partir de l'amplitude et de la largeur des phases des courbes. Ils sont reportés sur un cadran cartésien et apparaissent comme différentes courbes, avec un menu qui permet d'offrir une stratégie de visualisation de l'ECG pour un intervalle donné. Le schéma employé peut être utilisé pour mettre en évidence différents types d'anormalité.
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*Keywords:* Electrocardiogram (ECG); Heart rate; Visualization; PQRST wave; Arrhythmia

*Mots clés :* Électrocardiogramme (ECG) ; Rythme cardiaque ; Onde PQRST ; Arythmie

## 1. Introduction

The last two decades have witnessed the recognition of a significant relationship between the autonomic nervous system and cardiovascular mortality, including sudden cardiac death [1–4]. Experimental evidence for an association between propensity for lethal arrhythmias and signs of either increased sympathetic or reduced vagal activity has spurred efforts for the development of quantitative markers of autonomic activity.

Electrocardiography is the recording of the electrical activity of the heart. Monitored by placing sensors at defined positions on chest and limb extremities of the subject, electrocardiogram (ECG) is a record of the origin and propagation of the electrical action potential through cardiac muscle. This signal is representative of cardiac physiology, useful in diagnosing cardiac disorders. The state of cardiac health is generally reflected in the shape of ECG waveform and heart rate. It may contain important pointers to the nature of diseases afflicting the heart. However, bio-signals being non-stationary signals, this reflection may occur at random in the time scale.

The ECG signal is usually recorded in magnetic tapes (holter recording), a digitized version of which can be stored

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in computers. However, the above details are not clearly perceptible through casual observation of the raw signal on the screen. Therefore, it is convenient to analyze and classify the signal using a computer, and display the results for clinical observation. Cardiac disorders are currently diagnosed from electrical heart signal recordings (electrocardiograms or ECG) by identifying subtle variations in the twelve channels (leads) of ECG waveforms. To the trained eye the subtle variations in different channels can be correlated (in the mind) to indicate different cardiac disorders.

The regular normal heartbeat is generated by electrical impulses arising from the sinus node located in the right upper chamber (atrium). The heart muscle must be activated electrically before it can initiate its mechanical function of pumping the blood out of the heart into the body. The sinus node creates these spontaneous electrical impulses at a rate of 60–100 per minute at rest. This is manifested as a regular pulse at the same rate when one puts the finger on the front of the wrist to feel the pulsation of the artery. The sinus node can increase or decrease its rate in relation to various needs such as reduction in the rate during sleep to as low as 40 per minute, and increase to as high as 150 per minute during physical activities. The sinus node thus regulates the pumping of the heart in a rhythmic fashion and this is the normal rhythm of the heart. An arrhythmia is an abnormal rhythm outside of the normal variations mentioned above. In essence, an arrhythmia is an abnormal rhythm or beating of the heart. It can be due to abnormal function of the sinus node or it can arise from other areas which normally do not initiate electrical impulses. The arrhythmia can be manifested as rapid and regular heart beats (tachycardia), rapid and irregular heartbeats (fibrillation), slow heart beats (bradycardia), and abnormal extra beats occurring before the anticipated normal beats in a periodic fashion (premature contractions). Arrhythmias originating from the upper chambers (above the ventricles) of the heart are called supraventricular arrhythmias. Those originating in the lower chambers (ventricles) are called ventricular arrhythmias. The symptoms caused by arrhythmias depend on the type, severity, duration and frequency of arrhythmia. The severity of symptoms is also related to the underlying function of the heart. A young person with a normal heart may tolerate the arrhythmia very well but an older person with some associated heart disease will tolerate the same arrhythmia poorly. The arrhythmia-related symptoms are variable and include palpitations (feeling of rapid beats), weakness, fatigue, chest discomfort, shortness of breath, dizzy spells, and in the most severe form as actual loss of consciousness and cardiac arrest.

The efficacy of diagnosing diseases is therefore highly dependent on the level of training or expertise of the clinician, since the waveform variations are subtle. By using signal processing methods, these variations can be observed more clearly.

The QRS complexes and ventricular beats in an electrocardiogram represent the depolarization phenomenon of the ventricles that provides the useful information about their behavior. Beat detection is the important step in the ECG processing and analysis. For morphological analysis this is the reference for detection of other ECG waves and parameter measurements. Rhythm analysis requires classification of QRS and other ventricular beat complexes as normal and abnormal. Real-time ventricular beat detection is essential for monitoring of patients in critical heart condition.

Proper recognition of beats is impeded by power-line interference, electromyogram noise and baseline drift which are often present in the ECG signal. In long-term monitoring, electrode impedance can increase considerably resulting in very low signal-to-noise ratio, which can make detection practically impossible in a single lead. Therefore, usually two or three leads are used for monitoring [5].

Several studies to classify various cardiac arrhythmias have been reported [6–8]. A number of techniques have been used for identification of arrhythmias including correction waveform analysis [9], time-frequency analysis [10], complexity measures [11], and a total least squares-based Prony modeling algorithm [6]. Different features are extracted from the ECG for classification of ventricular arrhythmias including QRS and ST segment based values, heart rate, spectral features, AR coefficients, complexity measures and nonlinear measures [6–8].

In the past, Udupa et al. have proposed a set of algorithms for interactive visualization, manipulation, and measurement of large three-dimensional (3-D) objects (e.g. heart) on general-purpose workstations [12]. They have addressed the use of these algorithms for visualizing medical data. Morikawa et al. have investigated the feasibility of four-dimensional electrocardiography (4-D ECG), a new display in which the vector loop was rotated and scanned along a timed axis to overcome the shortcomings of vector cardiography (VCG) [13]. The P wave delineation score, signifying good agreement with the intraobserver and interobserver variability, was significantly higher in 4-D ECG than those in the orthogonal leads or those on the transverse and frontal projections. Acharya et al. have proposed a novel visualization technique to detect the various cardiac abnormalities easily [14]. They have assigned different colors for different types of arrhythmias and assigned a grid for each ECG beat. Later this technique is modified to sector graph, to visualize the entire 24 h of data, with an additional facility to peruse the details of chosen intervals [15]. Such techniques of 'software visualization' have been used in the past, to display programs, program artifacts and program behavior [16]. Paul et al. have proposed a scalogram plots to scrutinize the coordinated atrial activity during ventricular fibrillation [17,18]. He et al. have proposed the 3-D image of cardiac bioelectric source distribution from body-surface electrocardiograms [19]. Cardiac electrical sources were modeled by a current dipole distribution throughout the entire myocardium, and estimated by using the Laplacian weighted minimum norm (LWMN) algorithm from body-surface potentials. The approach we have taken is to combine the information from different ECG channels (leads) to construct an electrical dipole known as the cardiac vector which varies over time. This vector summarizes the electrical activities of the heart in the time domain.

In our work, we have considered the following types of cardiac arrhythmias [20–22]. The heart rate (beats per minute) is given by the expression

$$\mathrm{HR} = \frac{60}{t_{r-r}} \quad \text{(bpm)} \tag{1}$$

### 1.1. Types of cardiac conditions

#### 1.1.1. Normal sinus rhythm (NSR)

All P waves upright, rounded and similar in size and shape. P wave exists for every QRS complex. Each P wave is the same distance from the QRS complex – less than 0.20 seconds. All QRS complexes have the same size and shape and points in the same direction. Each QRS is at the same distance from the T waves and the duration of the QRS complex is 0.10 seconds or less. Heart rate is varying 60–100 bpm and is rhythmic.

#### 1.1.2. Ectopics

Preventricular contraction (PVC), bigeminy and couplets etc. fall in this type of abnormality. QRS complex is widened and not associated with preceding P wave. T wave is inverted after PVC. It is often followed by a compensatory pause. In couplets, there are two consecutive PVCs. In bigeminy, there is PVC after every other NSR [23–26].

#### 1.1.3. Atrial fibrillation (AF)

In atrial fibrillation, sinus rhythm does not occur. Instead, multiple "wavelets" of electrical impulses travel randomly through the atria, leading to random activation of different parts of the atria at different times. The walls of the atria quiver due to the tissues of the right and left atria are not stimulated to contract in an organized manner. Though this rhythm is irregular (when examined more closely), it appears to be regular.

#### 1.1.4. Complete heart block (CHB)

The heart rate will be usually between 30–35 bpm. P waves generated from the sinus node are not conducted to the ventricles because of block at the AV node. The P waves are indicated but show no relation to the QRS complexes. Since no impulses are conducted, the ventricular rate becomes dependent on spontaneous ventricular depolarizations. Severe symptomatic bradycardia with HR = 20–40 bpm are observed as the ventricles are depolarized by a ventricular escape rhythm.

#### 1.1.5. Right bundle branch block

Bundle branch block (BBB) is a disruption in the normal flow of electrical pulses that drives the heartbeat. Patients with BBB are generally without symptoms unless the disease is severe enough to cause a complete infranodal A–V block and very slow heart rate. In RBBB, the right bundle branch no longer conducts electricity. Therefore, as the electrical impulse leaves the HIS bundle, it enters left bundle branch only, and is carried to the left ventricle. Then, from the left ventricle, the electrical impulse finally makes its way to the right ventricle. As a result, the two ventricles no longer receive the electrical impulse simultaneously. First, the left ventricle receives the electrical impulse, then the right. The ECGs of RBBB have increased duration of the QRS complex (> 0.12 s) because it takes the right ventricle so much longer to depolarize [20–22].

#### 1.1.6. Left bundle branch block

In LBBB, the opposite occurs. Here, the left bundle branch no longer conducts electricity. The electrical impulse thus enters right bundle branch, and is carried to the right ventricle. From there, it finally spreads to the left ventricle. Once again, the two ventricles no longer receive the electrical impulse simultaneously. First, the right ventricle receives the electrical impulse, then the left. Technically for left bundle branch block to be present, the QRS duration is 120 ms or greater, there is poor R wave progression in V 1 thru V4 and the T wave vector is in the opposite direction to the QRS vector (T waves are inverted in I, V5 and V6).

#### 1.1.7. Left ventricular hypertrophy (LVH)

LVH is the abnormal thickening of the myocardium (muscle) of the left ventricle of the heart. While LVH itself is not a disease, it is usually a marker for disease involving the heart. Disease processes that can cause LVH include any condition that increases the after load that the heart has to contract against, and some primary diseases of the muscle of the heart.

Causes of increased after load that can cause LVH include aortic stenosis, aortic insufficiency, and hypertension. Primary disease of the muscle of the heart that cause LVH are known as hypertrophic. Left ventricular hypertrophy produces abnormally large leftward electrical forces that travel toward the left precordial leads V5–V6 and away from V1. P wave is wide and notched in leads I, II and V4 and V6 and biphasic P wave in leads V1 and V2. QRS axis is usually normal but maybe deviated to the left which is greater than $-30°$. Deep S waves are present in lead III and are 20 mm or 2.0 mV or more in lead III. T wave inversion is present in lead I and "downsloping" ST segment depression.

## 2. Materials and method

The 2D cardiac vector in the frontal plane may be reconstructed from the signals obtained from any two limb-lead ECGs, that is any two from leads I, II and III.

For this investigation, the digitized limb-lead ECGs are obtained from Physiobank's *Physikalisch-Technische Bundesanstalt* database. This database contains the full 12-lead digitized ECG data of healthy and nonhealthy patients, including those suffering from cardiac illnesses such as bundle branch block, myocardial infarction and cardiomyopathy. There is a header file, which describes clinical summary within each patient folder.

### 2.1. Preprocessing

Preprocessing involves the following:

- re-sample digitized signals to a sampling rate of 512 Hz. Original signal had a sampling rate of 1000 Hz;
- pass the data through a low pass filter with a cut-off frequency of 15 Hz to remove unwanted high frequencies present in ECG signal;
- apply a high pass filter with cut-off frequency 0.3 Hz to suppress baseline wander present in ECG signal;
- as the low pass filter is of second order and has a lower dB, a band-reject filter of cut-off frequencies 50 and 60 Hz to

suppress the power-line interference noise, as the low pass filter is of second order and has a lower dB;

- similarly, the high pass filter is also of second order and has lower dB, a median filter was used to extract baseline wander of the processed ECG signal, and then subtract it from the processed ECG signal to effectively remove all baseline wander.

### 2.2. Cardiac vector representation in 2-D

In this work, we attempt to investigate the construction and display of the cardiac vector over time. The display is limited to a 2-D representation of a cardiac vector although the cardiac vector is actually a 3-D quantity. The 2-D representation of a cardiac vector is more manageable as we acquire foundational knowledge to cope with the intricacies of displaying the 3-D vector.

By studying the shape of the path traversed by the cardiac vector, the observer can deduce, how the heart depolarizes and repolarizes. Cardiac disorders distort this path into different shapes and different orientations.

There are three distinct features of the ECG that the clinician uses:

- the P wave – which corresponds to the depolarization of the atrial muscles as the atria contracts and pumps blood into the ventricles;
- the QRS complex – which corresponds to the depolarization of the ventricular muscles as the ventricles pump blood out of the heart;
- the T wave – which corresponds to the repolarization of the ventricular muscles as the ventricles relax and gets ready for the next contraction.

In general, as the heart beats, the cardiac vector forms three distinctive loops that correspond to the three features. Their shapes, that is the size and orientation of the loops, and their locations in the display quadrants of the loop are all useful indicators that can be used to determine any cardiac disorder.

Take, for example, the cardiac disorders known as BBB; whether it is the RBBB or the LBBB is not readily discerned by looking at the ECG waveforms. But they can be determined more easily by studying how the QRS complex loop traverses the display quadrants. Taking Fig. 1 as an example, we can see that the QRS complex rotates through the display quadrants (observable when display is dynamic).

If the rotation crosses from the fourth quadrant to the first, it means the right ventricles depolarize first before the left ventricles. This is indicative of the LBBB. On the other hand, if the rotation crosses from the first quadrant to the fourth, it means the left ventricles depolarize first before the right ventricles. This is indicative of the (RBBB).

The way the heart depolarizes is reflected in the type of cardiac abnormality. This cardiac vector graphs may yield more information and provide an increase in accuracy in the detection of heart disease as compared to conventional methods.

The user-friendly interface allows the observer to easily choose which beat (ECG complex) out of the ECG recording

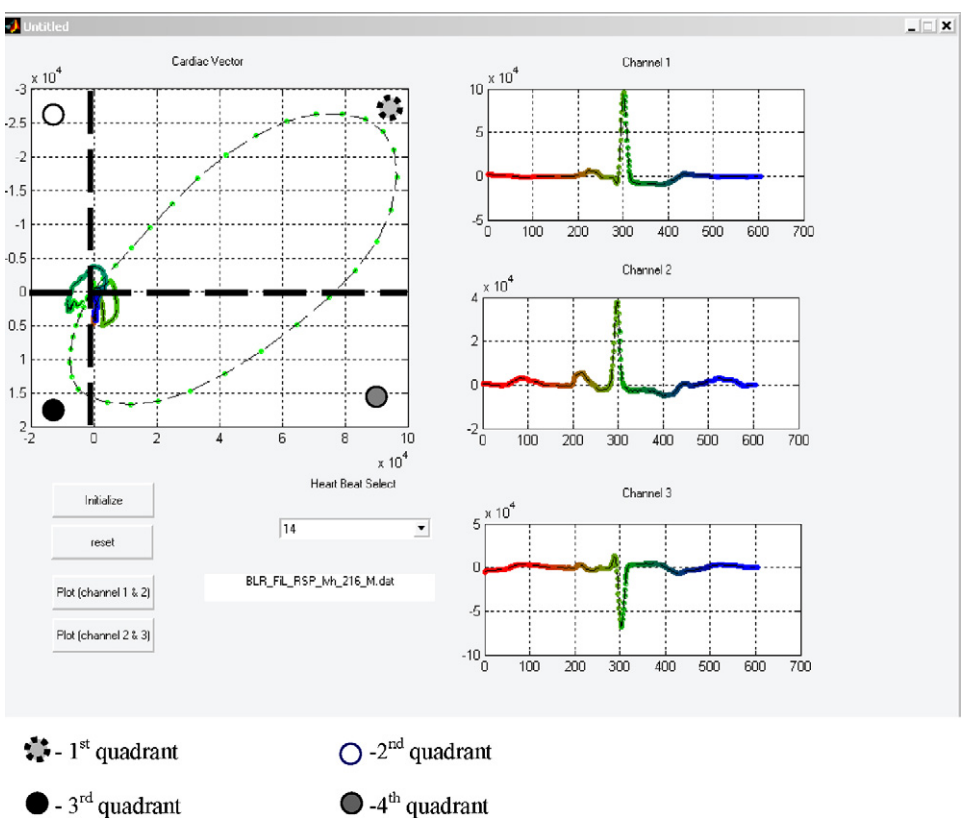


Fig. 1. Patient with left bundle branch block.

to plot out the cardiac vector. Distinctive variations in the shape of the cardiac vector appear for different cardiac disorders.

### 2.3. Lead selection for reconstructing of the cardiac vector

Two similar methods are used to reconstruct the cardiac vector. The methods essentially differ in the choice of using different limb leads. The first method utilizes the leads I and II, while the second utilizes leads II and III. The cardiac vector derived from either method yields similar plots with minor variations in magnitude, which is expected because of the different signals used. The method to select is dependent on which leads have better signals.

### 2.4. Mathematical derivation of the cardiac vector

The mathematical analysis for the reconstruction of the cardiac vector is obtained from the Eindhoven triangle formed by the limb lead ECGs, as shown in Fig. 2. The instantaneous cardiac vector $M_R$ is represented by the arrow in the middle of the triangle. The projections of the cardiac vector upon the leads I, II and III are represented by $M_1$, $M_2$ and $M_3$ respectively (Fig. 3).

#### 2.4.1. Case 1: Cardiac vector from Lead I and Lead II

The instantaneous angular displacement $\theta$ of the cardiac vector from lead I is given by:

$$\theta[n] = \tan^{-1}\left(\frac{((M_2[n]/M_1[n]) - 1/2)}{0.866}\right) \tag{2}$$

where $n$ is the sample number,

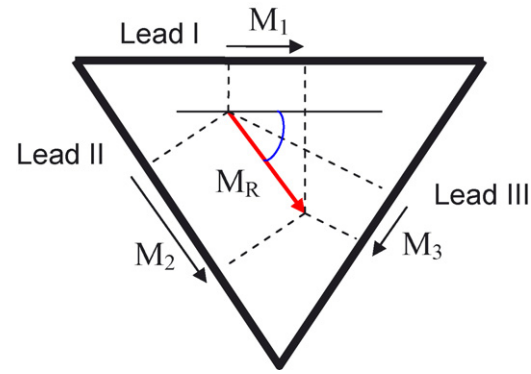


Fig. 2. Eindhoven's triangle.

Table 1
Algorithms for cardiac vector formula, constraints of the formula

| Condition ($n$ refers to the sample number) | Result of $\theta[n]$(degrees) |
|---|---|
| If $M_1[n]=0$ and $M_2[n]>0$ | 90 |
| If $M_1[n]=0$ and $M_2[n]<0$ | 270 |
| If $M_1[n]<0$ | $\theta[n]+180$ |
| If $M_1[n]>0$ and $M_2[n]<0$ | $\theta[n]+360$ |
| If $M_1[n]>0$ and $M_2[n]>0$ | $\theta[n]$ |

The instantaneous magnitude of the cardiac vector is thus:

$$M_R[n] = \frac{(M_2[n] - (M_1[n]/2))}{(0.866 * \sin\theta[n])} \tag{3}$$

Table 1 is needed to keep the equations (2) and (3) operational since computers are unable to cope with infinity values, which would occur for tangents of 90° and 270°.

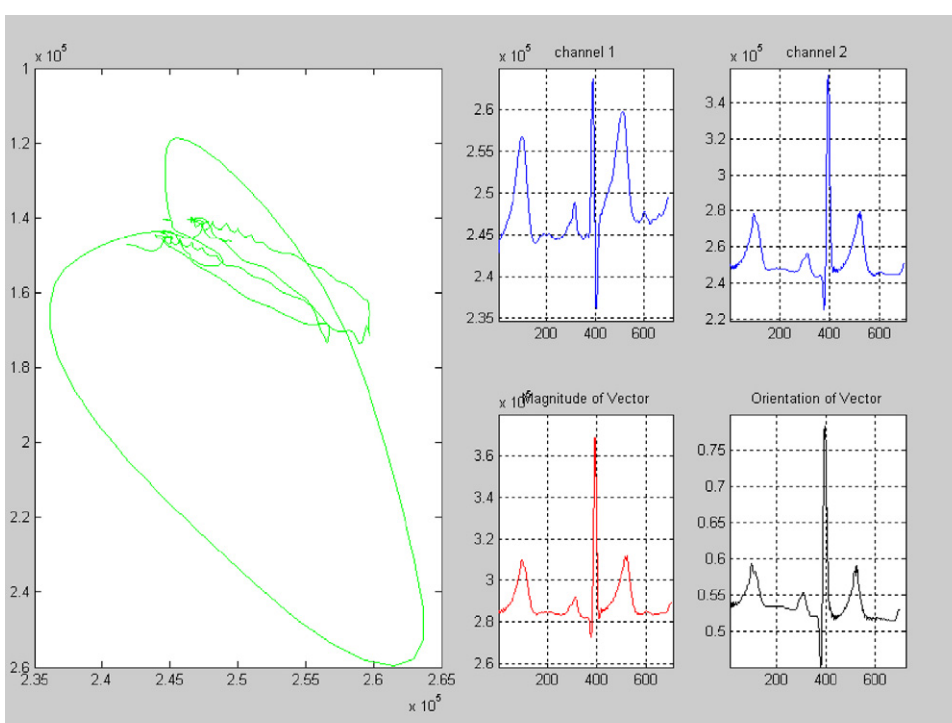


Fig. 3. Plot of channel 1 and 2 cardiac vector with magnitude and orientation.

The actual computation of the cardiac vector has the following steps.

*2.4.1.1. Step 1: Calculation of magnitude and orientation.* Each sample is used to calculate the orientation $\theta$[n] and the magnitude $M_R[n]$ from equations (2) and (3).

*2.4.1.1.1. Step 2: Conditions of matching.* Each sample of leads I and II is checked for the conditions in Table 1. For example, if $M_1[n] = -10$ (i.e. <0) and $M_2[n] = 10$ (i.e. > 0), the actual orientation $\theta[n]$ is thus derived by adding 180° to the calculated orientation in equation (2).

2.4.1.1.1.1. Step 3: Plotting of the cardiac vector.2.4.1.1.1.1. Step 3: Plotting of the cardiac vector The result of $\theta[n]$ and $M_R[n]$ are plotted in polar format to yield a graphical cardiac vector plot over a short period of time (typical cardiac period is 1 s).

*2.4.2. Case 2: Cardiac vector from leads II and III*

The formulae for deriving the cardiac vector from leads II and III is similar to that of deriving from leads I and II, if we note that the orientation of leads II and III has an angular offset of 60° to that of leads I and II.

Thus we can write:

$$\alpha[n] = \tan^{-1}\left(\frac{((M_3[n]/M_2[n]) - 1/2)}{0.866}\right) \quad (4)$$

$$M_R[n] = \frac{(M_3[n] - (M_2[n]/2))}{(0.866 * \sin(\alpha[n]))} \quad (5)$$

where the variable $\alpha$ is the angular orientation with respect to lead II.

The angular orientation $\theta$ with respect to lead I is therefore:

$$\theta = \alpha + 60^\circ$$

Table 2 is used to determine the actual orientation $\theta$ for the cardiac vector. $\alpha$ has to be processed to $\alpha_1$ before it can be used to calculate the actual orientation $\theta$.

*2.5. Plotting of the cardiac vector*

*2.5.1.1. Step 1: Initialization of data*

The user first enters the ECG beat number (HB) to be analyzed in the GUI provided. For example, the physician want to visualize the cardiac vector of 22nd ECG beat, then he needs to enter 22 in the Heartbeat select bar.

This initialization will generate a digitized data of the PQRST waveform corresponding to the HB number and becomes the basis for computing the cardiac vector.

Table 2
Algorithms for cardiac vector formula, constraints of the formula

| Condition ($n$ refers to the sample number) | Result of $\alpha_1$ (degrees) | Result of $\theta$ (degrees) |
|---|---|---|
| If $M_2[n] = 0$ and $M_3[n] > 0$ | 90 | 150 |
| If $M_2[n] = 0$ and $M_3[n] < 0$ | 270 | 330 |
| If $M_2[n] < 0$ | $\alpha[n] + 180$ | $\alpha_1[n] + 60$ |
| If $M_2[n] > 0$ and $M_3[n] < 0$ | $\alpha[n] + 360$ | $\alpha_1[n] + 60$ (if $\alpha_1[n] < 300$)<br>$\alpha_1[n] - 300$ (if $\alpha_1[n] > 300$) |
| If $M_2[n] > 0$ and $M_3[n] > 0$ | $\alpha[n]$ | $\alpha_1[n] + 60$ |

Table 3
Algorithms for phase extraction

| Condition | Algorithm |
|---|---|
| Start of a phase | If $\theta$ is increasing and mR is increasing, then it is start of phase. |
| End of a phase | If $\theta$ is decreasing and mR is decreasing, then it is end of phase. |

*2.5.1.1.1. Step 2: QRS detection.* In this step, an ECG signal with 550–600 samples corresponding to the heartbeat number (HB) selected will be displayed. The QRS detection algorithm will run automatically to detect the R peaks [23–26].

*2.5.1.1.1.1. Step 3: Processing of the ECG signal.* The ECG data samples within the identified window are used to compute the instantaneous orientation [$\theta$] and magnitude of the cardiac vector [$M_R$].

2.5.1.1.1.1.1. Step 4: Cardiac vector display.2.5.1.1.1.1.1. Step 4: Cardiac vector display The orientation [$\theta$] and magnitude of the cardiac vector [$M_R$] are plotted using these two polar co-ordinates.

Different segments of the PQRST waveform are plotted using different colors to indicate the progression of the cardiac vector. The plot starts with red, changing colors to green and finally to blue as it ends.

*2.6. Features extraction*

The phase information of each features like P wave, T wave and QRS complex are detected first. The interval between two successive QRS complexes is defined as the r–r interval ($t_{r-r}$ s), is determined by the QRS algorithm [23–26]. After the detection of QRS complex, the window is shifted behind by 100 samples to search for the interval of P wave. Then another window is shifted ahead (from QRS complex) by 170 samples to search for T wave interval. The onset and the offset of the peaks are required to detect the P wave and T wave intervals. The onset of T wave, defined as the T point, is found by first locating the T wave peak which is the maximal absolute value, relative to the isoelectric line, between J + 80 ms and R + 400 ms. The onset of T wave, the T point, is then found by looking for a 35-ms period on the R side of the T wave, which has values within one sample unit of each other (Table 3).

## 3. Results

In these plots, there is QRS width, P interval and T interval is plotted. The plot shows the resultant of lead I and lead II waveforms. And the resultant is shifted by an angle of $\theta$ between two samples.

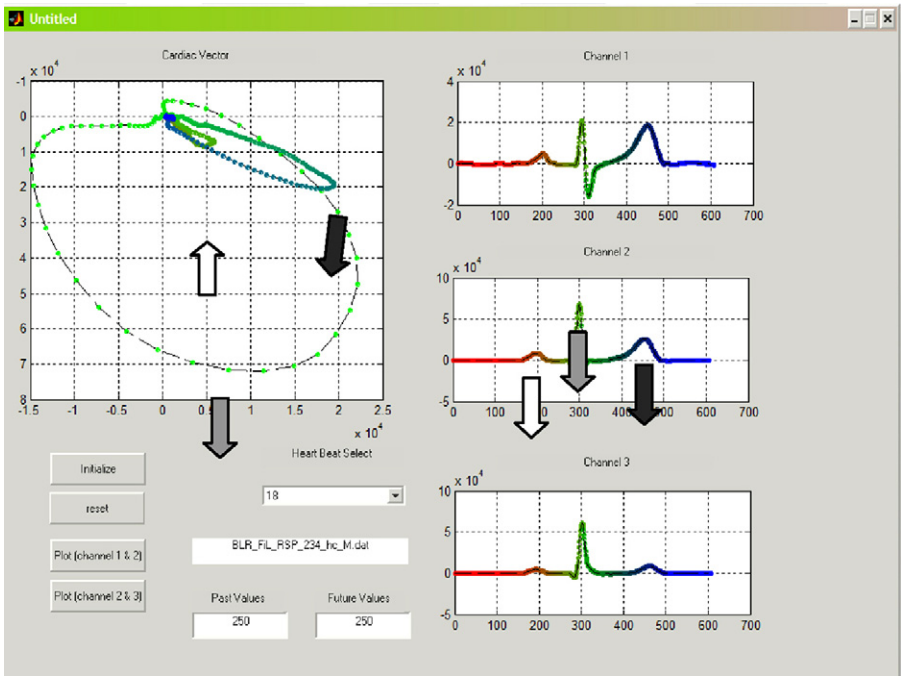


Fig. 4. Vector cardiogram of normal sinus rhythm.

The Fig. 4 shows the cardiac vector of a normal sinus rhythm. The biggest loop (in grey arrow) represents the QRS complex and the second largest loop (in black arrow) represents the T wave. In this figure, P wave loop, which is the smallest loop (white arrow), has been partially overlapped and circumferenced by the T wave loop and so the cardiac vector plot may appears to have only two loops at a casual glance. In general, there will always be three cardiac loops for a normal sinus rhythm. The P wave loop is smaller in magnitude and has the same orientation as the larger T wave loop. It is more readily viewed by selectively choosing the portion of the ECG to plot just the P wave.

The Fig. 5 shows the cardiac vector of a complete heart block. There are distinctive differences from a normal sinus rhythm. The QRS loop and the T wave loop are not centralized, an effect of the deep S wave in the QRS complex. The QRS loop also has its own unique shape.

Vector cardiogram corresponding to the left bundle branch Block beat is shown in Fig. 6. The curve corresponding to the

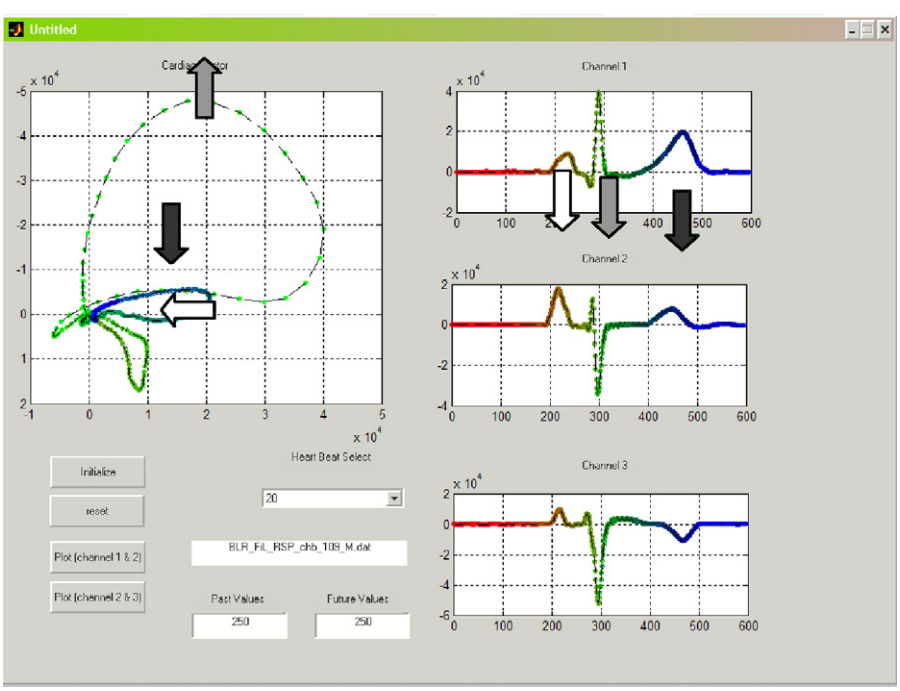


Fig. 5. Vector cardiogram of complete heart block.

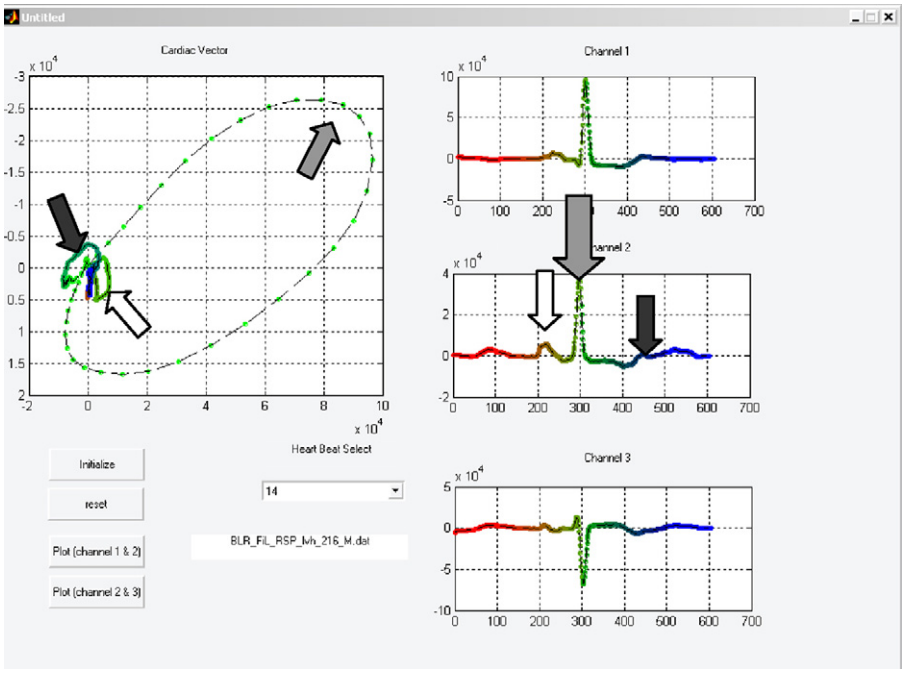


Fig. 6. Vector cardiogram of left bundle branch block.

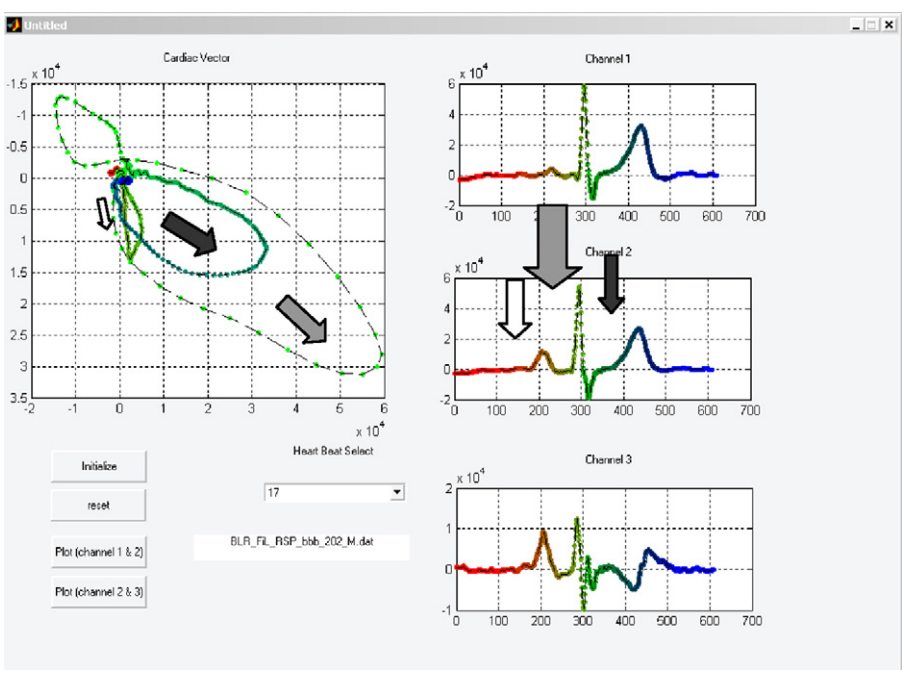


Fig. 7. Vector cardiogram of right bundle branch block.

QRS complex duration is oriented towards first quadrant, P wave and T wave duration is clearly seen.

The Fig. 7 shows the vector cardiogram plot of the right bundle branch block. The curve corresponding to the QRS width duration is shifted to the 3rd quadrant.

In the vector cardiogram corresponding to atrial fibrillation (Fig. 8), the curve corresponding to the QRS complex is oriented towards 3rd quadrant and the curve corresponding to the P wave is absent.

## 4. Discussion

In this paper, we have proposed the visualization of cardiac disorders using cardiac vector plot. In our work, we have achieved early stages of recognizing these disorders by looking at the loop patterns for each heartbeat. For example, the orientation and the magnitude of the loop indicate the polarization of the heart and it helps in the detection of left or right bundle branch block.

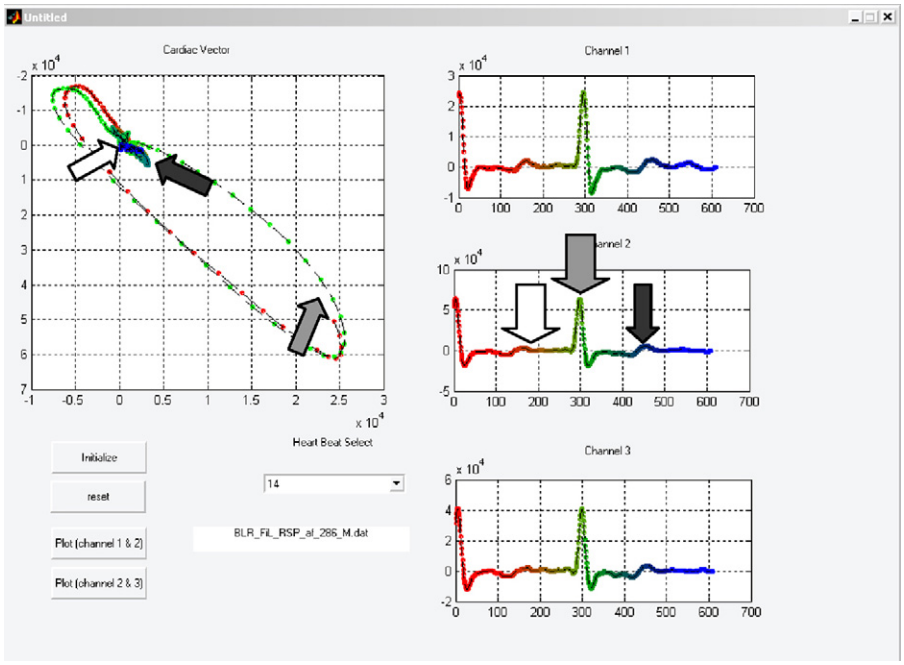


Fig. 8. Vector cardiogram of atrial fibrillation.

ECG signals are essentially nonstationary signals; they display a fractal like self-similarity. Disease symptoms are highly subjective and it occurs at random in the time scale. Therefore, the physician may have to observe the patient for long intervals to diagnose the affliction. The ECG may contain important information about the nature of disease, both in its shape and duration between its salient peaks and troughs. The actual values and variation of these parameters are not obvious for a casual survey of the waveform upon the screen. However, to (study and) pinpoint anomalies in voluminous data collected over several hours is strenuous and time consuming. However, the problem can be effectively handled by using computer for analysis and display of salient features of the signal. In this study, features are extracted from the ECG and plotted as new plot using these features.

In Fig. 4 (normal subject), the QRS loop is a nice oval shape occupying the 3rd and 4th quadrants, showing the polarization of the heart during the QRS interval. In addition, it is in the same orientation with the T loop. This indicates that, the QRS complex and T wave are not inverted from each other. Fig. 5 (complete heart block), has an abnormal QRS loop indicating the improper polarization of the heart. The QRS loop and the T loop are also not aligned in the same orientation signifying that, one of them is inverted. In this case, the QRS loop occupies the 1st and 2nd quadrants and so the QRS loop is inverted. Lead I, of both Figs. 4 and 5, looks similar. It is very difficult to detect any cardiac disorder just by inspecting the ECG at lead I. Similarly, other abnormalities are indicated clearly in their shapes and orientations.

More information about the ECG signal can be obtained by converting its features in to the cardiac vector plots. Using huge database and with more features these results can be further improved.

## 5. Conclusion

A visual display platform for the study of subtle changes in the ECG data is developed. This display technique is useful for identification of abnormalities, which would considerably reduce the tedium involved in the manual scrutiny of data. In this technique, the features extracted for each type of abnormality, has unique plot. The orientation and shape of these of curves will vary from disease to the other. This technique does not need extensive computations and is simple to operate. The platform is expected to be of significant help to study the effects of drugs, treatment and therapy.

## 6. Hardware and software specification

The program to eliminate the baseline wander and extract the different features: P interval, QRS width and ST interval are written in MATLAB.

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